AutoML Insights: quebec_housing_sales_v2.csv

AutoML Analysis & Business Insights Report

Prepared For: PredictIQ Enterprise User
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Platform: PredictIQ Studio AutoML Engine

1. Dataset Profile & Summary

The AutoML engine parsed the dataset **quebec_housing_sales_v2.csv**. Below is an overview of the dataset dimensions and data quality metrics.

Metric	Value
Dataset Name	quebec_housing_sales_v2.csv
Total Row Count	4,000
Total Column Count	13
File Size	208.2 KB
Data Quality Score	**100/100**

2. Applied Data Cleaning Steps

The following data cleansing and preprocessing pipeline steps were executed prior to training:

- Impute Missing Values: lot_size_sqft
- Drop High Missing Column: renovation_year
- Encode Categorical Columns
- Remove Outliers (IQR Method)
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3. AutoML Model Comparison

A total split validation was run on multiple algorithms. The leaderboard below displays performance metrics. Algorithms are ranked based on their validation score.

Model	R^2 Score	MAE	RMSE	CV Score	Time
XGBoost	83.98%	45324.9766	60647.2070	83.14%	1.658s
Gradient Boosting	77.33%	54869.0555	72137.6239	78.72%	0.407s
Random Forest	76.98%	54648.3750	72703.1823	78.41%	0.094s
Linear Regression	48.48%	85649.5638	108759.2127	50.68%	0.001s
Decision Tree	47.71%	78186.2500	109568.0896	53.27%	0.027s

4. Best Model Deep-Dive

The best performing model is **XGBoost**. It was selected automatically based on optimization targets for 'sale_price'.

5. AI Business Insights & Recommendations

This section outlines tactical business findings derived from model metrics, weights, and feature importance.

Executive Summary

We performed automated machine learning (AutoML) on the dataset 'quebec_housing_sales_v2.csv' with the goal of predicting the target column 'sale_price'. The problem was identified as a regression task (estimating a continuous numerical value). After evaluating multiple models, the XGBoost model was selected as the optimal predictor with a R^2 score of 84.0% and Mean Absolute Error (MAE) of 45324.9766.

Data Quality Review

Quality Rating: Excellent (Score: 100/100). The dataset has minimal missing values and duplicates. Data cleaning steps successfully handled remaining discrepancies. Columns with high missing rates or high variance were normalized, and outliers were analyzed to improve stability.

Primary Business Drivers (Feature Importance)

The analysis indicates that the most influential features driving predictions are: living_area_sqft, sale_year, city. These parameters hold the highest weights or split-gains, making them key leverage points for understanding model behavior.

Actionable Recommendations

- Focus business strategy adjustments around the primary drivers (living_area_sqft, sale_year, city).
- Continuously monitor model performance as new data flows in to verify that 'sale_price' matches predictions.
- Implement data ingestion validation checks to maintain high data quality scores.

Model Strengths	Model Weaknesses
• Highly optimized preprocessing pipeline handled missing values and encoding automatically. • The XGBoost model is robust against overfitting and offers balanced metrics.	• Rule-based fallback was used instead of Gemini LLM. Standard summaries might lack contextual domain explanations. • If some features are highly correlated, individual feature importance could be slightly distributed.
Possible Bias & Risks	**Suggestions for Improvement**

• Class imbalance or distribution skew in target 'sale_price' can skew model outputs. Regular review is advised.	• Provide a Gemini API key in Settings to unlock deep, contextual generative business analysis. • Add more domain-specific features or collect more rows to further boost validation metrics.